



Bridging the gap: Do community needs assessments enhance public participation in transportation planning?

Miriam Pinski^{*}, Lauren N. McCarthy

Shared-Use Mobility Center, Los Angeles, CA, USA

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ABSTRACT

Traditional avenues of public input in the transportation planning process require residents to have the time, knowledge, and capacity to participate. Those who don't – often lower-income and other disadvantaged groups – are left out. To encourage more broad-based participation, some states and localities have funded additional community-based, grassroots-organized transportation studies to explicitly engage with underrepresented populations. However, the effectiveness of such studies is an open question. This study analyzes the results of a pilot program in California to determine whether community studies, or community studies, include the disadvantaged groups they intended to include, and whether they offer additional benefits or costs compared to traditional public participation processes in transportation planning. The California Air Resources Board funded a series of community transportation needs assessments under the Clean Mobility Options pilot program that were intended to include historically underrepresented groups in identifying their transportation needs and potential solutions. We examine responses from those programs' resident surveys, community engagement strategies, and findings from the needs assessments' final reports, to identify factors that predict successful outreach. We also compare public participation methods in community needs assessments with those in long-range transportation plans. Although we are unable to ascertain whether the community studies led to any changes in local transportation systems, we do identify the conditions under which additional engagement encourages traditionally hard-to-reach communities to participate in transportation planning, as well as when it does not. We conclude this paper with recommendations for when additional studies can be useful, and alternative policies to address equity gaps in the transportation planning process.

1. Introduction

Traditional public participation in the transportation planning process – typically public meetings and opportunities to review and comment on plans – has faced criticism for failing to adequately achieve satisfactory public participation (Innes & Booher, 2004). Critics argue that such procedures advantage people with the resources (time, knowledge, ability) to participate and exclude low-income and disadvantaged populations in particular (Creger et al., 2018). In response to these concerns and growing advocacy for equity in transportation planning (Creger et al., 2018; Massey, 2020), state and federal policymakers have recently prioritized more community-based participation.

In California, policymakers within the state's Air Resources Board proposed a more inclusive and bottom-up approach to public participation by funding community transportation needs assessments (CTNAs). These needs assessments intend to empower eligible project

areas to engage historically underrepresented and underserved populations in identifying transportation gaps and finding solutions tailored to their local needs. Whether the community studies do so, however, is an open question. Legislative mandates and community study grants present an opportunity for residents and local stakeholders to have a more prominent voice in the transportation planning process. Yet scholars of public administration and planning continue to debate the merits of community participation in the policymaking process (Webler et al., 2001).

In this paper, we explore the benefits and limitations of traditional public participation methods in transportation planning and investigate the role of community transportation needs assessments in enhancing inclusivity and community-centric planning. We use the terms community studies and needs assessments interchangeably. We analyze CTNA applications, surveys, and final reports from across the state of California, focusing on examining survey respondent populations,

^{*} Corresponding author.

E-mail address: miriam@sharedusemobilitycenter.org (M. Pinski).

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engagement efforts, and factors that predict successful survey outreach. We also compare needs assessment methods with community engagement methods used by public agencies in traditional long-range transportation plans. Based on our analysis, we present potential conditions when engagement can improve public participation and better serve disadvantaged communities in transportation planning, as well as when it does not.

2. The benefits and limitations of public participation in planning

Community participation is considered a best practice in planning decision-making; advocates often tout resident participation to balance competing interests and offer disadvantaged constituents a voice (Creger et al., 2018; Massey, 2020). Long-range transportation plans, an established method by which regional and state agencies create a comprehensive vision for their region's transportation future, are required under federal law to incorporate public involvement efforts. Government agencies typically rely on public hearings and opportunities for the public to review and comment on proposed projects and plans to satisfy legally-required participation (Innes & Booher, 2004).

Advocates justify public participation procedures on the grounds that they offer policymakers a chance to learn from residents about their preferences and local knowledge that can improve a policy; they legitimate government decisions; and they advance equity (Innes & Booher, 2004). Inclusive planning procedures might produce what Arnstein and scholars who continued in her tradition consider "citizen power" (Arnstein, 1969; Slotterback & Lauria, 2019). By directly giving members of the public the ability to influence policy decisions as well as build social capital and trust within their community, public engagement gives residents "real power" and more capacity to affect change (Slotterback & Lauria, 2019).

Public participation is intended to counterbalance the government's authority to set policy that directly affects residents. Advocates for local input in transportation planning often cite the federal highway construction and urban renewal projects of the 1950s and 1960s that displaced more than 1 million Americans, residents that were disproportionately low-income and Black and had little political influence over the process (Karas, 2015; "The Interstates and the Cities," 2002). By the early 1960s, amid growing public backlash against urban highway construction and the civil rights movement, Congress began to mandate a more inclusive decision-making process. The federal Highway Act of 1962, the Civil Rights Act of 1964 and subsequent transportation and environment justice laws expanded participation in land use and transportation policy, with an increasingly explicit goal of remedying the negative consequences from earlier closed-off and destructive highway construction projects (Handy, 2008; Karner & Marcantonio, 2018).

More recently, in 2021, President Joe Biden signed the Bipartisan Infrastructure Law, which continued to mandate public involvement in transportation planning. The law specifies that metropolitan planning organizations (MPOs) must incorporate public involvement throughout the process of developing their metropolitan transportation plans and transportation improvement plans. MPOs must work with the public to develop a formal public participation plan, and states, too, are required to document their public process. Planning agencies must provide adequate public notice, enable the public to review and comment on key decision points, and offer multiple formats to ensure accessible participation.

Yet, there is a growing call among planning scholars and activists that those open participation procedures polarize the planning process rather than productively engage residents, that they do not close longstanding gaps in representation, and that they fail to meaningfully improve social equity in both the transportation planning process and mobility outcomes (Manauagh et al., 2015). Additionally, public participation itself may simply tokenize community members, rather than

actually incorporate their needs.

Public participation has often been characterized by a representation gap, where participants tend to be older, white, and relatively affluent, not fully representing the broader community demographics (Einstein & Glick, 2017; Manville & Monkkonen, 2021). This representation gap between residents who participate in community meetings and the overall community remains even after participation shifts online (Einstein et al., 2023). Representation in local issues can tangibly impact local investments; if the people who attend a meeting to discuss a new public transit route never envision themselves using it, but constitute a minority among a broader potential transit-riding population, their voices may have outsized influence. Even well-meaning residents may not accurately reflect the broader community's interests.

Participation in transportation decisions tends to attract residents seeking an avenue to oppose, rather than amend or support, proposals (Gordon, 2022). As with other land use changes such as housing development, transportation projects often have concentrated costs and diffused benefits. The minority of stakeholders for whom those costs are acutely felt are, unsurprisingly, most motivated to speak out in a public forum. Take, for example, a proposed protected bike lane that might encourage more people to bike or prevent a fatal collision but removes street parking. Adjacent business owners might worry that they can no longer provide as much parking to their customers, and some residents active in local politics might actively oppose projects that remove space from drivers. In contrast, the beneficiaries of the bike lane are broad (cyclists in and around a neighborhood who can enjoy a few blocks of added protection) and hypothetical (a potential victim that avoided a hypothetical crash). The project might spur a small group of opponents and advocates into action, but it is likely not worth the typical beneficiary's time to participate in a lengthy process for a marginal improvement in their area's street network.

Establishing a consensus around any policy is never easy, but especially when communities are diverse. Further, scholars argue that true consensus only can be claimed when all interests have been explored and an effort has been made to satisfy those concerns (Innes, 2004). Not every member will benefit from every policy, and thus some will inevitably oppose funding programs they don't derive value from themselves (Demsas, 2023). For example, adding a protected bike lane can offer active residents a safer, more affordable alternative for short trips than driving. Still, it will not benefit people with limited mobility, or it can be perceived as an inconvenience to drivers.

However, true consensus is often not the goal of public participation, as public participation seeks to identify the extent to which public opinion has influence over an idea brought forward by a public agency, rather than explore all possibilities (Thomas, 2003). In light of these limitations, some planners propose alternative methods to address gaps in both transportation planning processes and outcomes. Needs assessments offer one potential avenue for more inclusive policymaking. In general, a needs assessment is intended to directly involve residents, often including local stakeholders in developing the assessment and forming recommendations. Typically, needs assessments focus on analyzing the existing transportation system and areas for improvement by engaging stakeholders and the public with focus groups, interviews, and especially surveys. More recently, some needs assessments have incorporated alternative methods of engagement such as photovoice journals, or walk and bike safety audits. The specific format of a needs assessment informs who participates and what feedback they provide. A focus group or public meeting might encourage people willing to speak publicly, and discourage others from speaking openly. A survey, while more confidential, limits responses to the questions and answer format selected by the survey designer.

Other planning scholars advocate for more transparency in transportation plans as to how projects are prioritized, and to weight equity considerations in the process (Wennink & Krapp, 2020). Alternatively, the role of local participation could shift from the public meeting or community study to the ballot box. Citizens can – and increasingly do –

vote on local options sales taxes to directly influence the types of projects that get funded (Hannay & Wachs, 2007). They can also vote for elected officials to represent their transportation policy preferences. More people vote in state elections than attend public meetings, and thus elected officials may make choices that better reflect the median voter. Rather than concentrate veto power among the small sliver of the population that participates in local public meetings, state representatives could assume more authority over broader regional transportation matters. Moving decisions from the local to the state level could thus better democratize planning investments.

3. Background

We focus our analysis on 24 CTNAs conducted across the state of California as a part of the Clean Mobility Options Voucher Pilot Program (CMO).

This program was selected for further analysis for a few reasons. California offers an influential case study as it is a large state whose experiments with transportation policy reverberate across the country. California has, moreover, long embraced public participation in transportation planning. As early as the 1950s, California introduced laws like the Ralph M Brown Act and the California Public Records Act, guaranteeing the public's right to attend and participate in meetings of local legislative bodies. More recently, it established the CMO pilot program, part of the California Climate Investments (CCI) initiative in California. CTNAs fall under the umbrella of CMO programs, which are broadly intended to improve clean transportation access and increase zero-emission mobility choices, particularly in disadvantaged and low-income communities.

The CMO CTNA program offers a case study of community studies that are explicitly targeted at including who the state considered underrepresented in the planning process. While needs assessments are not unique to transportation, they are growing in popularity. Though CTNAs are part of a voluntary voucher program and not mandated, they are an emerging tool in a policy area that is expanding in part due to state and federal government requirements to facilitate public participation in long-range transportation plans. CTNAs go beyond the traditional public participation processes in transportation planning by directly funding community-based organizations to carry out their own studies. The state intended for CTNAs to be grassroots, community-led efforts, and for CTNAs to offer disadvantaged residents an opportunity to gain a stronger voice in transportation planning. As such, only eligible disadvantaged communities, as well as low-income tribal and affordable-housing communities, could apply. Once accepted into the program, awardees were required to submit quarterly progress reports, and upon completion of the program, fill out an exit survey consisting of 31 questions that asked for such things as satisfaction of the technical assistance and other aspects of the program. Awardees were also required to submit final summary reports of their work. Survey data was recommended but not required to be submitted to the administrator team.

The first funding round of CTNAs launched in June 2020, with \$1.15 million available for eligible communities. Vouchers worth up to \$50,000 were provided to improve transportation options over a period of nine months, between 2020 and 2021. Awards were not competitive but awarded on a first-come first-served basis. The 24 awardees are made up of a mixture of public agencies, nonprofits, private consultant groups and American Indian tribes. Some needs assessments were led by public agencies, others by community-based organizations or tribal groups; in some cases consultants co-led the process. Organizations had varying levels of expertise and previous experience with community studies. All awardees had access to technical assistance but varied in how much they utilized it.

Each CTNA application required applicants to describe their target community and why that community would benefit from an additional study. Once awarded, awardees were then required to conduct a resident survey to assess existing travel behavior and gaps in their transportation

network, analyze at least two publicly available data sets (such as American Community Study surveys) and two community engagement events. CTNA awardees submitted a final report detailing their efforts and findings to complete the program. The program did not include an official mechanism by which final CTNA reports, once finished, would influence transportation plans or funding decisions. Awardees could, however, use their CTNA to apply for another voucher-based Clean Mobility Options program, the Mobility Voucher Program, to fund a shared mobility pilot program. The Mobility Voucher Program, which funded zero-emission microtransit, scooter-share, bikeshare, and car-share programs, required a completed needs assessment in the application.

4. Data and method

Our study addresses a few key questions related to public participation. First, we want to better understand who specifically organizations that conduct community studies aim to include in their studies. Second, we want to determine whether the data they collect do, in fact, represent their intended community. Third, we seek to compare needs assessment outreach methods to public participation outreach efforts used in traditional transportation planning. Fourth, we aim to identify under which conditions community studies are most successful in achieving participation goals.

To answer our questions, we relied on a range of sources from CTNA awardees. We conducted a content analysis of CTNA application materials to identify whom each needs assessment sought to include in their study, and analyzed their final reports for their chosen methods and final recommendations. Using the final reports, we examined the types of recommendations awardees suggested (e.g. changes to land use, improved safety, additional shared mobility services) and the degree to which those recommendations offered specific, rather than general, proposed changes (e.g. general improved safety versus proposing a specific safety intervention). We also examined awardees' survey responses they filled out at the end of the CTNA program to better understand the challenges and benefits of the program, as well as their community survey responses to determine whether they ultimately included their intended community members. The surveys allowed us to identify the types of strategies and characteristics of organizations that were most and least successful in including their target community in their surveys. Finally, we conducted additional analyses for five CTNA project areas to compare their process and outcomes with their region's broader long-range transportation plan. We use the terms "target community," "target population," and "population of interest" interchangeably to refer to the segment of the project area's population that the awardee aimed to prioritize in their CTNA.

5. Analysis

5.1. Reaching target populations

Despite identical grant amounts and timelines, awardees garnered vastly different sample sizes for their surveys. The sample sizes ranged from nearly 3,000 responses to 12; the median number of responses was 168 responses. Survey sample sizes can vary based on the characteristics of the project area (its population size and geography), the target population demographics (e.g. older adults or people who do not speak English as a first language), and the type of organization conducting a needs assessment. In evaluating survey success, we measured the aggregate number of responses and adjusted them for population size, and compared whether the profiles of survey respondents matched those whom the awardees intended to target in their application.

Needs assessment awardees were not required to capture samples that represented their entire project area's population. They could instead define their community from within the broader project area, so long as the selected populations lived or worked within the eligible

project area, they could collect data as it fit the process. This is not unusual for needs assessment methodologies, as targeted populations within a census tract often differ from the overall demographic characteristics.

In their applications, awardees listed the target population groups they planned to engage in their outreach efforts. In some cases, projects defined their communities broadly as all residents in their project area. Other projects planned to focus on a subset of residents in their project area, such as farm workers or students, faculty, and staff of a community college. Table 1 lists these groups alongside the number of awardees that named them in their applications and their proportion compared to all awardees.

The most common identified groups included low-income (92 percent), Hispanic/Latino/a people (listed in 83 percent of applications), non-English speakers (75 percent), people without access to a private vehicle (58 percent), people without a driver's license (54 percent), immigrants (54 percent), and people without regular access to the internet (50 percent). Many of these groups overlap, such as immigrants and non-English speakers, or people without driver's licenses and those without access to a private vehicle.

5.2. Did the surveys achieve Representation?

The needs assessment pilot program was designed to include traditionally underrepresented populations. Who, exactly, has been underrepresented in the planning process is not defined by the state with any specificity. That seems to be intentional; while the term “disadvantaged” refers to places that fall under a clearly defined range of the CalEnviroScreen Index, which incorporates a number of pollutant exposures and socioeconomic risk factors, the term “underrepresented” has no such criteria. A 2022 report by the California Department of Water Resources advocated that the term “underrepresented” remain broad so as to enable state agencies to be as inclusive as it wants in determining who is eligible for state funding (Agustinez & Brown, 2022). In the case of needs assessments, it is thus left to awardees to define who is underrepresented in their planning process.

Noticeably, most awardees did not attempt to identify key demographic questions to ascertain whether the people they surveyed matched their intended population group. For example, several awardees stated their targeted population was agricultural workers in their needs assessments, but none asked whether respondents were farm workers. Awardee C, a county government, stood out in that they provided demographic data on the residents of their project areas alongside their county averages, and later compared the demographics of their survey respondents to those in their project areas to establish whether

they had a representative sample. However, their sample was too small (just eight responses) to elicit any statistically significant analysis.

We selected a range of projects to identify whether the people who completed surveys represented the target population profile the awardee identified in their application (see Table 2). The projects represent a mix of urban and rural areas, awardee types (a nonprofit, consultant group, public agencies, and a tribe), and target populations. Our selection was limited to needs assessments that listed specific population groups in their application and included questions in their surveys to identify those members.

Many awardees (75 percent) intended to overcome language barriers to public participation and specifically target non-native English speakers. The extent to which those awardees were able to include them depended on the language and outreach effort. Spanish speakers were much more likely to participate in translated surveys than speakers of other languages. Nonetheless, some awardees in areas with large Spanish-speaking areas received few Spanish survey responses. Awardee D, for example, collected 102 survey responses; 87 percent of their respondents identified as Hispanic/Latino/a, but just four completed the survey in Spanish.

Outreach to people who spoke neither Spanish nor English was particularly challenging for awardees. One awardee found that despite offering their survey in an indigenous language, Mixteco, many Mixteco speakers were fluent in Spanish and used the Spanish survey instead. Another awardee went to great effort to translate their survey into Hmong, but received no Hmong responses. This may reflect the fact that many people who do not speak English or Spanish live with relatives or near family or friends who can translate for them. It may also reflect the particular challenge in reaching non-English speakers who may not just be linguistically isolated but physically isolated from broader society as well.

Identifying groups to target for engagement was much more straightforward when membership to a group is clearly defined, such as an American Indian tribe (Awardee A), and the project area is small and circumscribed, such as a college campus. The most successful organizations also cast the broadest net in terms of their target population. For example, Awardee A's target community included tribal members who lived or worked on the reservation. Very few people in their project area did not meet their criteria, and, unsurprisingly, every survey participant matched their target profile. Outreach was also easier when the leading organization had experience conducting outreach (e.g. focus groups or developing government plans), such as a government agency that works on regional plans, or an organization with strong ties to other community groups or public agencies, like a nonprofit or university that routinely collaborates with other community groups.

5.3. Challenges and successful approaches to survey outreach

Because many awardees chose not to ask key demographic questions, we cannot know whether they achieved representative results. We can, however, better understand why they chose not to ask those questions. In some cases, awardees intentionally chose not to ask certain questions out of privacy concerns. Citizenship and driver's license status, for example, could identify a person's legal status in the country if not properly anonymized. Some awardees also worried that respondents would discontinue the survey if they were asked personal questions they were uncomfortable with, such as their household income. One awardee, Awardee B, gathered just 40 responses, only half of which included responses to demographic questions.

A second explanation is that awardees insufficiently prioritized demographic questions. Awardees often found that people were unwilling to answer lengthy surveys with over 40 questions. Many awardees shortened the surveys to encourage participation, cutting demographic questions rather than transportation-related ones.

A possible related explanation is that many of the implementing organizations had a skill gap or limits to research capacity. Even though

Table 1
Populations Studied in Clean Mobility Options Transportation Needs Assessments.

Target Population	%	Number of Assessments
Low-Income	92 %	22
Latino/a	83 %	20
Non English Speakers	75 %	18
Carless	58 %	14
No Driver's License	54 %	13
Immigrant	54 %	13
No Internet	50 %	12
Unemployed	46 %	11
Tribe	42 %	10
Farm Workers	33 %	8
Students	29 %	7
Older Adult	21 %	5
Community Worker	21 %	5
Undocumented	17 %	4
Single Parent	17 %	4
School Faculty/Staff	13 %	3
Homeless	8 %	2
Businesses	4 %	1

Table 2
Transportation Needs Assessment Target Populations and Survey Results.

Awardee	Target Population	Survey results	Sample Size	Awardee type	Region type
A	Tribal residents, who are primarily working families who commute or work in Big Pine.	100 % Native American	176	American Indian tribe	Rural
B	Lowest-income households in the community, homeless people, the elderly, children, the physically and mentally challenged, those with poor health	22 % 65 or older; 9 % under 18 years old; 43 % households earning under \$50 k	40 but for each demographic question, only 22/23 responded, 8 completed long survey, 94 short surveys	CBO	Urban
C	Lower levels of education and income, undocumented community members that cannot vote, and residents with a language other than English spoken at home.	60 % annual household income of less than \$49,999		Public agency	Urban
D	Spanish speaking households, seniors, residents without driver's licenses, undocumented populations, agricultural workers, people living below the poverty line, residents without internet access, and people without access to a personal vehicle.	92 % at least 1 vehicle in household; 32 % older than 50; 87 % Hispanic/Latino; 44 % primarily language Spanish, 56 % English; 66 % annual household income less than \$25,000; 29 % report no regular Internet access	35	Public agency	Rural
E	Unemployed and underemployed residents	94 % employed	116	Nonprofit with consultant group	Urban

awardees had access to technical assistance, they may have underestimated both the importance of demographic variables, and the work involved deploying physical surveys to get a desired sample size. Survey participation depended in part on the organization's research capacity and their willingness to put resources into outreach. Some community-based organizations, for example, relied on snowball survey sampling techniques that resulted in mostly responses from other people already active in community organizations. In contrast, organizations that routinely conduct community surveys and public outreach, such as planning agencies or consultant groups, encountered less of a learning curve in conducting a community study.

Awardees that included private consultants yielded more responses; CTNAs with a consultant had a median sample of 194 responses compared to 78 for CTNAs without a consultant on the project. After adjusting for population size, awardees with consultants on staff gathered three times as many responses as those without. We surmise that consultants had professional experience in and could devote more resources to community outreach efforts compared with other organizations whose members needed to divide their time between the needs assessment and other ongoing work projects.

A third explanation may be that the natural and built environment characteristics play a role in outreach: an organization conducting a needs assessment in a rural area where jobs and residents are dispersed and people have inconsistent access to the internet may have more trouble engaging the population than one working in a project area smaller in size, where residents live near each other. Whereas a tribal group may be able to reach every member in its project area, due to its small group size and close network ties, a less known or less trusted organization might struggle to establish trust among residents and collect many responses.

People who live in places that are more spread out can be more difficult to reach, particularly with surveys taken in-person or online surveys in places where internet access is limited. Aggregate survey responses were, unsurprisingly, lower in rural project areas than urban ones; the median survey sample size in a rural area was 54 responses compared with 190 in urban areas. After adjusting for population size, differences in survey sample sizes were no longer meaningful (a *t*-test shows no statistically significant difference between rural and urban, before and after accounting for tribal projects; sample sizes were very small, however). Organization and population characteristics may instead play a bigger role in survey response rates than the built environment.

The type of outreach efforts that organizations chose also plays a role in survey turnout. Many groups specifically attended events or mailed their surveys to places they were confident they would reach their

community of interest. One group sent its surveys to every resident of an affordable housing complex. Another found a captive audience among farmworkers waiting upwards of two hours at a regular food distribution event. But sending copious flyers and other advertisements about a survey was not a sure sign of successful outreach. For example, Awardee D distributed flyers advertising their survey through utility bills and mailed flyers to households that their local Community Services District identified as still needing to receive utility bills. The awardee, a rural transit agency, nonetheless received very few responses for the survey.

In-person surveying efforts may also have encouraged more people to respond to and complete surveys. Convenience is key. Asking community members to attend something they otherwise would not was, some awardees found, less effective. One awardee found that initiating events that required participants to go somewhere they ordinarily would not, or to spend a significant amount of time, dissuaded turnout:

"The majority of our community engagement events with the public piggy-backed off pre-organized events scheduled by school sites. While our event during a Back-to-School night at X Elementary reached a high number of parents, we weren't able to get deeper feedback due to the event logistics. At our focus group events at a monthly parent night we got incredible feedback from participants but only reached a total of seven parents. In the future, we need to conduct our own outreach for events or join onto an event where having deeper conversations is already built into event schedules."

With the right outreach efforts, an awardee working in a rural community could garner just as high a survey sample as an awardee in a densely populated urban area. Awardees that hired locally, or included local volunteers, could also better identify where their target populations were most likely to congregate. Some awardees in rural areas met churchgoers after services. Another awardee met residents where they were and when they had time to participate at a food distribution event, where approximately 300 farmworkers waited upwards of two hours in line.

Our findings highlight that needs assessment organizers often encounter the same challenges engaging hard-to-reach population groups as traditional planning processes do. Reaching people who are socially isolated, perhaps because of physical disability, or who are uninterested in participating in studies out of personal preference, distrust, or time constraints, is a challenge even for community-based organizations. People who are unemployed, for example, may be difficult to identify and include in survey outreach. Awardee E, for example, wanted to engage with unemployed and underemployed residents, but nearly everyone that took their survey (94 percent) was employed. Another organization, Awardee B, wanted to prioritize children, but

only nine percent were under 18.

Yet, organizations were able to successfully engage their target communities under specific conditions. Those who targeted small populations, such as small American Indian tribes, allowed for focused and in-depth engagement. Additionally, organizations with experience in outreach, such as consultant groups with expertise in conducting focus groups or developing government plans, played a crucial role in effectively reaching and involving various stakeholders. Furthermore, successful engagements often involved organizations with strong ties to other community groups or public agencies, like nonprofits or universities with a history of collaboration, enhancing community trust and cooperation. Lastly, conducting outreach events in locations where residents naturally gathered, such as attending pre-arranged parent/teacher events, proved to be an effective strategy for engaging residents where they were most comfortable, facilitating their active participation in the needs assessment process.

5.4. Comparing public participation approaches in needs assessments to traditional planning

We find that public participation processes under existing long-range transportation plans are often similar, if not far more extensive, than community engagement methods conducted for needs assessments (see Table 3). Each plan included a survey, public meetings and workshops (often co-hosted with community-based organizations), and targeted outreach to non-English speakers, tribal areas and/or low-income and communities of color. Some plans went beyond public meetings and public comment by offering interactive activities like collaborative mapping, demonstration events, and focus groups. Most public participation documents documented public feedback, but few offered a demographic analysis of who actually participated. Of the five regional

transportation plans, one listed proportions of where respondents lived.

6. Discussion: Weighing process versus outcomes

An open question – and ongoing debate – is to what extent community studies improve the planning process. There are two related components to this question: whether such studies make the planning process more representative, and whether the findings from community studies influence planning decisions and outcomes.

As to the first question, our findings are mixed. In certain cases, awardees included their target populations in survey outreach. An experienced planning agency in some cases had the advantage of experience in surveying and public outreach, and were more likely to ask relevant demographic questions in their surveys. But just as likely residents might not respond to surveys from a public agency they distrusted or had little personal interest or connection to. In contrast, a small community-based organization might better engage with residents they had pre-existing ties to. But small organizations might also be inexperienced in conducting surveys or transportation research, be less likely to ask relevant demographic questions, and, depending on their organization’s capacity, generate a small and skewed sample.

Might the type of data collection change representation? As we described, when we analyzed CTNA survey responses, we found many target population groups underrepresented in survey responses. More often than not awardees did not emphasize who actually responded to surveys. Some awardees also conducted additional focus groups and interviews, but were even less likely to track their participants’ demographic information with those methods. We cannot, therefore, evaluate whether these alternative forms of participation led to the widespread inclusion of target communities. One awardee remarked in their final report that while they did not collect data on the people who participated in their focus groups, they appeared to be primarily white and older. Other studies that examine demographic representation in public meetings find that participants are more likely to be white and older compared to their community’s demographics (Einstein et al., 2023). Surveys like the US Census and travel behavior surveys like the National Household Travel Survey also historically under-count low-income groups, racial/ethnic minorities, and immigrants (Blumenberg & Smart, 2010; Landis, 2022; Stempowski, n.d.). These studies suggest that unrepresentative samples in needs assessment surveys may reflect broader obstacles to participatory planning than specific to needs assessment protocols.

We found that planning organizations conducting long-range transportation plans also tended not to track who participated in their outreach efforts. Some regional transportation plans outlined the metrics they would use to track their public participation. In one plan for a very populated county, the agencies planned to count the number of meetings held in “traditionally underrepresented communities,” the number of advertisements for public comment, surveys, and documentation for how public input was incorporated. The plan listed the following groups it considered “traditionally underserved and under-represented”: children and youth, individuals with access and functional needs, low-income communities of color, older adults or retired people, populations with limited English proficiency, and women and female-headed households. For the most part, however, public participation plans did not identify mechanisms to identify who participated, particularly demographics characteristics, and so agencies could not verify that their plans reached the audiences they intended to reach.

Yet, even if an awardee or agency is able to capture 100 percent of their population in their community engagement, the process is costly. Residents and community-based organizations may have limited time and resources to contribute. They may be even less likely to contribute if their previous experiences assisting with community studies did not lead to any tangible outcomes. Moreover, such studies constitute a significant information demand from residents who may have little background in land use and transportation policy. It may well be the case that the

Table 3
Public Participation Methods in Needs Assessments and Long-Range Transportation Plans.

Awardee	CTNA Method	Regional Plan Public Participation Process	Targeted Groups in Long-Range Transportation Plan
A	Workshop, survey, stakeholder phone calls	Online survey, requested feedback on RTP, public hearing	Tribal government
B	Survey, public forum, photovoice (1 person participated)	Public meetings, workshops co-hosted with CBOs, request public input, listening sessions, interactive activities (e.g. collaborative mapping), demonstration events, online and printed surveys	Non-English speakers, tribal councils, Asian, Latino and African American stakeholder groups
C	Surveys, workshops, interviews with residents	Online survey, public meetings, surveyed residents for preferred input method	Non-English speakers, tribal councils
D	Survey, focus group, community meetings	Public meetings/workshops, co-host workshops with CBOs, breakout sessions, online and printed surveys distributed at public meetings and transit hubs, focus groups	Low-income and minority neighborhoods, tribal governments, women, non-English speakers
E	Survey, interviews, focus groups with virtual reality component	Public hearings, informal workshops, co-hosted meetings with CBOs, surveys,	Youth, non-English speakers, Asian, Hispanic and Black stakeholder groups

planning process is thorough in attempting to include diverse community groups, but that residents themselves choose not to participate. A metropolitan planning organization, or a needs assessment awardee, may go to great lengths to include disadvantaged populations in their study, but those residents may have other priorities or limited time or interest to guide local transportation policy. Compensating participants, and offering virtual forums to participate, are considered best practices by some needs assessment proponents, and can go part of the way to including residents. But people only have so much time, and attention, to give in a day.

As to the second question – whether studies influence planning outcomes – additional research is needed. The outcomes of public participation are notoriously difficult to track (Rowe & Frewer, 2004). Agencies operate under constrained budgets, and are unlikely to try to ascertain the extent to which, for example, additional focus groups translated into changes in funding or, ultimately, people's travel behavior.

A community transportation study, by providing more resources to directly engage with residents, might offer useful insights for planners as to which transportation improvements are most popular or necessary. After analyzing the contents of all 24 needs assessment final reports for their recommendations, we found that the most commonly cited mobility barriers were high costs (88 percent), access to transportation (79 percent), and safety concerns (63 percent) (see Table 1 in the Appendix for a list of mobility barriers awardees stated in their applications and final reports). The findings and recommendations logically varied by geography and organization type. A rural project was less likely to propose bike lanes, and an organization that consults on electric carshare was more likely to propose a carshare system. Some assessments offered concrete and specific recommendations: a charging station should be placed at *this* library, a pedestrian light at *this* intersection. On the whole, however, the problems they highlighted – that driving was expensive, but alternative forms of travel dangerous or inconvenient – and the solutions they identified – flexible clean microtransit options, safer active transportation infrastructure – were bound by common themes and solutions.

Recommendations were practical and, in some cases, seemed to reflect awardee goals and expertise. Awardees were more likely to recommend zero emission shared vehicles (included in three-quarters of recommendations) than active transportation (58 percent recommended active transportation improvements). Since most project areas were car-oriented, vehicle improvements were a more immediate solution for mobility-constrained populations than active transport improvements like road diets and other land use changes, which tend to require longer-term investments and broader coalitions to implement. Moreover, most respondents (74 percent) planned to use their CTNA report to apply for the CMO mobility voucher program, which offers funds restricted to clean mobility fleet purchases and operations, and not street improvements that make walking and cycling more convenient and safer. Some awardees included bicycle advocacy groups or electric vehicle specialists who advocated for their own mode-specific service expansions. Others advocated for trips to certain locations; one awardee, for example, included a coalition of organizations committed to improving access to outdoor recreation. Though their survey results showed that the vast majority of surveyed residents had access to at least one household vehicle (95 percent), their survey questions seemed designed to encourage responses that would support their preferred mobility program. When asked “If local organizations offered free transportation to participate in their programming locally at locations like a state park and beach, would your family participate more frequently?” 88.5 percent of adult and 72.2 youth participants responded yes.

The value of additional community studies can be questioned if their findings could be more readily identified without the need for surveys and other costly outreach efforts. If recommendations are generally predictable, then perhaps data-driven approaches to identifying needs offer a far less resource-intensive process. We find that

recommendations from community studies tend to overlap with existing long-range transportation plans. Long-range transportation plans in every region – urban and rural – stressed the region's needs for safety, accessibility, and clean transportation for its disadvantaged, lower-income communities.

While the details of a community's recommendations matter, so does the process for what happens after the final report is completed. A community-based study might rigorously identify transportation gaps and propose solutions, but if those solutions are not approved by the powers that be, they may never take root. To actualize plans, some awardees would need to use additional resources to take their recommendations to the relevant authority – like their city council – and go through the required official processes to find funding and gain approval. Needs assessments face the same implementation challenges as other previous participatory planning efforts. In Sacramento, California, for example, planning experts, ordinary citizens and local elected officials co-developed regional plans to achieve sustainable transportation land use under a so-called “blueprinting” process. Yet scholars found that the long-term participatory planning effort ultimately led to very little change in the regional transportation system and land use patterns, and on net the region increased its greenhouse gas emissions (Niemeier et al., 2015).

7. Conclusion

Public participation is a required aspect of long-range transportation planning, and agencies may choose the simplest procedures to check legislatively-mandated boxes. Community transportation studies can provide additional resources to include resident voices in the planning process, but our findings show that they, too, are prone to similar challenges in incorporating disadvantaged population groups. Under our case study program, the CTNA awardees were required to conduct a community survey and two additional forms of community engagement, which were up to the awardee to select. We found that most engagement types were similar, if not less rigorous, than community engagement used in their project area's long-range transportation planning process.

Our analysis sheds light on the conditions under which public participation through needs assessments can enhance findings and identifies its limitations. By comparing the demographics of survey respondents to the populations of interest that awardees listed in their application materials, and identifying the types of outreach methods used by awardees that garnered the largest survey samples, we identified the methods that were most effective at encouraging broader community engagement. We found response gaps between target survey populations and survey users. We also found evidence that awardees tended not to ask pertinent demographic questions and that they encountered similar challenges that traditional planning agencies do when they try to address hard-to-reach populations.

Our findings suggest that needs assessment surveys can successfully address targeted populations under certain circumstances, particularly when the target population is small, when the organization is experienced in outreach, and when they leverage their connections with other community groups or public agencies. Locals, or people familiar with local events, could help facilitate broad-based participation. A relatively straightforward best practice for outreach was surveying residents in places they regularly gathered.

Done well, a community study has its benefits. They can offer organizations funding to reach out better, include underserved residents, and better democratize the transportation research process. Advocates argue that needs assessments create platforms for community engagement that establish trust between mobility providers and residents, and between community members and public agencies. Advocates also claim that needs assessments familiarize residents with both the planning process and with alternative forms of transportation.

They do, however, come with drawbacks. Needs assessments are not a one-size-fits-all solution for addressing underrepresentation in public

participation. In some cases, such goals could potentially be achieved more directly with existing public participation processes without additional study requirements. Community-based transportation needs assessments require time, money, and research resources. Those resources also represent opportunity costs: money that could be used to fund mobility interventions or reduce carbon emissions directly is instead directed towards additional studies. Moreover, public agencies must spend additional resources approving and incorporating needs assessment findings into their existing transportation plans.

We recommend that transportation agencies incorporate targeted outreach efforts that encourage broad-based participation into existing long-range planning practices, rather than create additional studies that generate more regulatory steps to fund, carry out, and then implement the resulting findings. To avoid redundant studies and future survey fatigue, we also recommend that organizations conducting assessments review existing studies before engaging the community with a new study. Before starting a new study, an organization should weigh the benefits a new study might offer a community with the additional costs (if any) they impose on disadvantaged populations. Agencies funding or requiring independent needs assessments should consider only requiring or voluntarily conducting them under distinct circumstances in which an additional study may be necessary, such as, for example, when a novel transportation technology is introduced, or when there is sparse existing data on the safety or accessibility issues in a region.

Our analysis shows that under certain conditions, participation expectations through needs assessments can be met. Still, suppose participation is ineffective in pushing outcomes that benefit disadvantaged communities. In that case, additional studies may expose a bigger flaw in the planning process. Future research addressing this should focus on understanding to what extent the independent needs assessment conducted under this or similar programs replicates existing long-range transportation plans or reports, and the longer-term outcomes and

impacts of independent needs assessments on transportation systems.

CRediT authorship contribution statement

Miriam Pinski: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lauren N. McCarthy:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table A1. Stated Needs in Applications vs. Found Needs in Final Report.

Mobility Barrier/Need	Stated in Application	Found in Final Report
Access	54 %	79 %
Safety	17 %	63 %
Knowledge	0 %	54 %
Cost	50 %	88 %
Health	25 %	17 %

Table A2. Final Recommendations.

	Percent Recommend	N
Bikes	45 %	11
E-Bikes	33 %	8
Carshare	63 %	15
E-Carshare	67 %	16
Carpool	29 %	7
Transit	46 %	11
Bus/Shuttle	58 %	14
Transit Routes	21 %	5
Transit Frequency	21 %	5
Scooters	17 %	4
E-Scooters	13 %	3
On-Demand Transit	29 %	7
Smartphone App	0 %	0
Partnerships	71 %	17
Zero Emission Vehicles	96 %	23
Energy Infrastructure	67 %	16
Safety	58 %	14
Street Design	63 %	15
Education	54 %	13

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